Wear — Elsevier Sequoia S.A., Lausanne – Printed in the Netherlands

EROSION BY A STREAM OF SOLID PARTICLES

J. H. NEILSON AND A. GILCHRIST
The University of Strathclyde, Glasgow (Gt. Britain)
(Received May 28, 1967)

SUMMARY

In this work, experimental results are given for the erosive action of a particle-laden gas stream on specimen materials which have widely different physical properties. The effects of particle shape, particle velocity and angle of attack are investigated and the parameters affecting particle deposition in the specimen surface are enumerated. A simple approach to a theoretical analysis of the problem is given and the relationships derived are used to correlate the experimental results. These relationships are used to predict the erosion–angle of attack characteristics for different specimen materials.

INTRODUCTION

The erosive action of fast-moving particles is of importance in many engineering applications. In the pneumatic transportation of solids, for example coal or grain, the enclosing ducts may suffer severe damage. In gas turbines, blade erosion by particles leaving the combustion chamber, may considerably shorten the life of the machine and there is evidence that the erosion by water droplets in the low pressure stages of stream turbines is a similar phenomenon. In rocket engines, part of the surface damage to the exhaust nozzle wall is attributed to the action of particles from the combustion chamber. On the credit side there are the useful applications of particle attack in shot peening, sand-blasting and erosive drilling of hard materials.

A wide variety of experimental data is available on different aspects of the problem [1–6]. In general, this material cannot be correlated because neither the velocity of the attacking particle nor the angle of attack was observed.

FINNIE[7–9] in an important contribution in this field likens the particle to the cutting edge of a tool which moves into the specimen surface causing plastic deformation of the material and removal of the debris so formed. He compared his theoretical results with experiments in which the particle velocity and angle of attack were strictly controlled. His correlation was good for ductile specimens at small angles of attack. His theoretical relationships, however, make no prediction for the erosion which occurs under normal attack on such materials. BITTER[10] also makes a theoretical analysis of the problem in which the type of wear analysed in FINNIE's work is known as cutting wear and is accompanied by plastic deformation of the surface material. In addition, however, BITTER allows for deformation wear which is associated with the repeated blows suffered by the specimen and which eventually cause cracking and

spalling of surface material. It is this deformation wear which accounts for the erosion at normal attack in ductile materials and which is not accounted for in FINNIE's analysis. BITTER's theoretical work is exhaustive and extremely intricate, accounting as it does for the elastic as well as for the plastic properties of the particle and specimen materials. The complexity of BITTER's final relationships and other factors prompted the authors to seek a simpler analytical solution to the problem of erosion which could readily correlate experimental data.

EXPERIMENTAL WORK

The apparatus used in the experiments is shown in Fig. 1. Air was expanded through a convergent two-dimensional nozzle and particles contained in a reservoir above the nozzle were allowed to flow through a restriction, to control the particle flow rate, before being introduced to the air stream at the nozzle inlet. Just upstream of

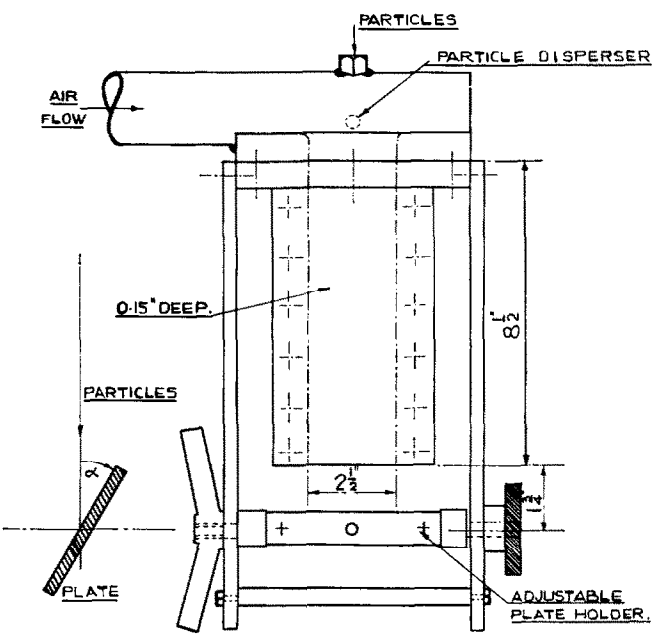


Fig. 1. Apparatus used for tests.

the nozzle inlet the slow-moving particles struck a rod which was arranged to give an even distribution of the particles across the inlet cross-section of the nozzle. The particles were accelerated by the expanding air and the efflux of air and particles were allowed to strike external plate specimens. These specimens were mounted on a plate-holder so that the angle, α, between the plane of the nozzle and the surface of the plate could be varied. The particle velocity at the point of impact on the plate surface was obtained using a photographic technique in which two exposures of the particles on one negative were obtained with a known time interval between them. In the experiments, aluminium, glass, perspex and carbon plates were tested, using glass spheres and angular aluminium oxide particles of different particle size as the eroding agent.

RESULTS

Figure 2 shows the change in weight of an aluminium plate at $\alpha = 90°$, when subjected to attack by increasing amounts of 210 μ aluminium oxide particles at various particle velocities. These results indicate that, initially, particles are deposited in the specimen and that the mass to be impacted to cause a given weight loss by erosion depends on the particle velocity.

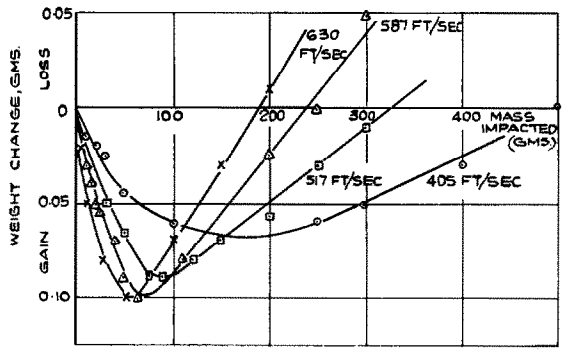
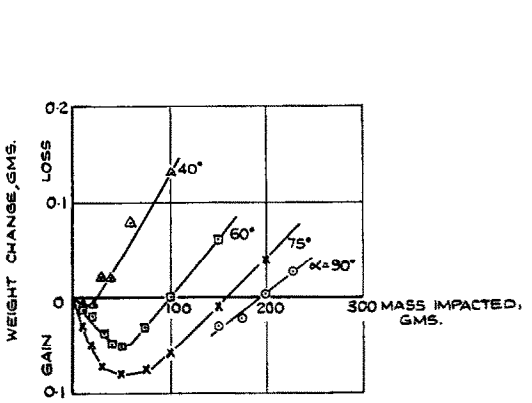


Fig. 2. Weight change *vs.* mass impacted for aluminium plates. $\alpha = 90°$, 210 μ aluminium oxide particles with various velocities.



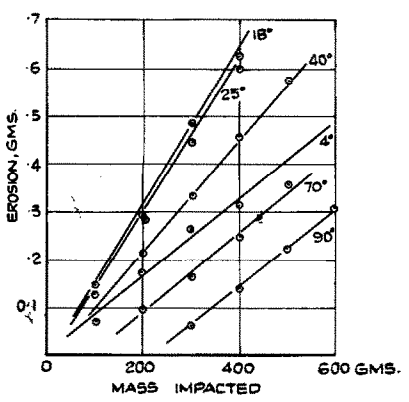


Fig. 3. Weight change *vs.* mass impacted for aluminium plates. 210 μ aluminium oxide particles at 630 ft./sec. Various angles of attack.

Fig. 4. Erosion *vs.* mass impacted results for aluminium eroded by 210 μ aluminium oxide particles at 424 ft./sec.

Tests, similar to those described above, were made in which the particle velocity was constant at 630 ft./sec and the angle of attack was varied. The results are shown in Fig. 3 and indicate that with decreasing angle of attack the mass to be impacted to cause erosion decreases. The results from several such tests showed that, once erosion is established, the weight loss per unit mass impacted is constant for a given angle of attack and particle velocity. The results for these linear portions of the erosion characteristics for aluminium plates attacked by 210 μ angular aluminium particles at 424 ft./sec are shown in Fig. 4. From these results the erosion per unit mass–

angle of attack characteristic shown in Fig. 5 was determined. The erosion– angle of attack characteristics for aluminium plates with the same particles at velocities of 354 ft./sec and 494 ft./sec were obtained in a similar manner and are shown in Figs. 6 and 7. In Fig. 8 the angle of attack characteristics for aluminium plates eroded by 475 μ diameter glass spheres at 260 ft./sec is shown. For these spherical particles no deposition effects were observed.

Figure 9 shows the characteristic for glass plates eroded by 210 μ angular

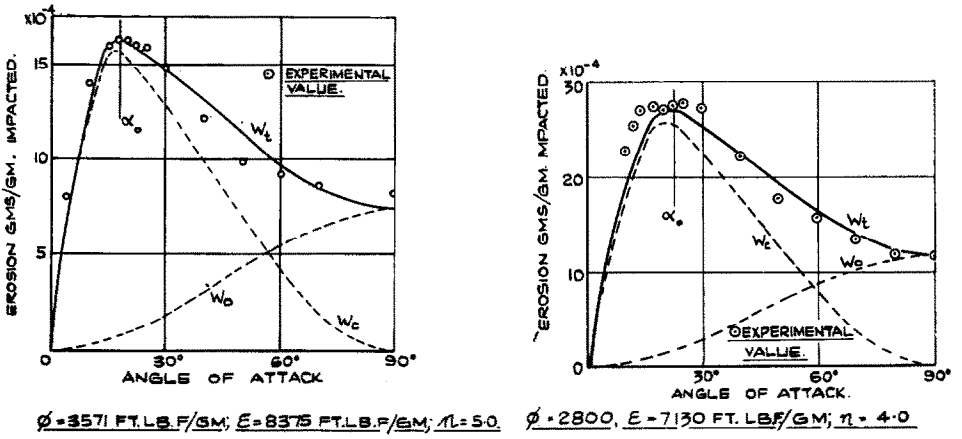


Fig. 5. Erosion *vs.* angle of attack characteristic for aluminium eroded by 210 μ aluminium oxide particles at 424 ft./sec.

Fig. 6. Erosion *vs.* angle of attack characteristic for aluminium eroded by 210 μ aluminium oxide particles at 494 ft./sec.

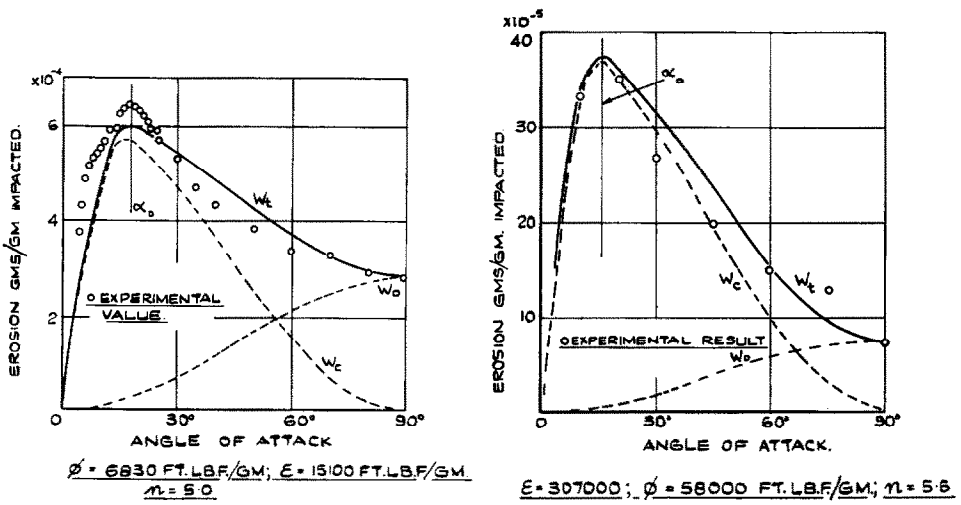


Fig. 7. Erosion *vs.* angle of attack characteristic for aluminium eroded by 210 μ aluminium oxide particles at 354 ft./sec.

Fig. 8. Erosion *vs.* angle of attack characteristic for aluminium eroded by 475 μ diameter glass spheres at 260 ft./sec.

aluminium oxide particles at 354 ft./sec and in Fig. 10 a similar result for glass plates and 297 μ aluminium oxide particles at 316 ft./sec is given. The characteristic for glass plates and glass spheres of 475 μ diameter at 260 ft./sec is shown in Fig. 11. It will be noted here that no erosion occurs at angles of attack less than 35°. For all tests using glass plates no deposition effects were observed.

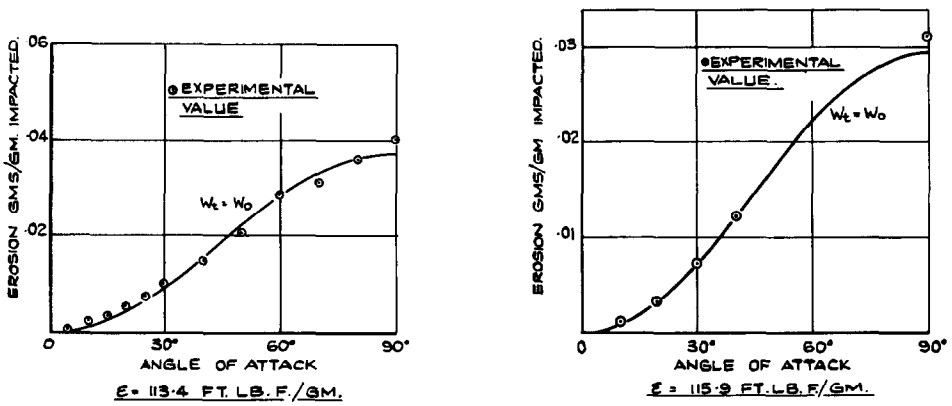


Fig. 9. Erosion *vs.* angle of attack characteristic for glass plates eroded by 210 μ aluminium oxide particles at 354 ft./sec.

Fig. 10. Erosion *vs.* angle of attack characteristic for glass plates eroded by 297 μ aluminium oxide particles at 316 ft./sec.

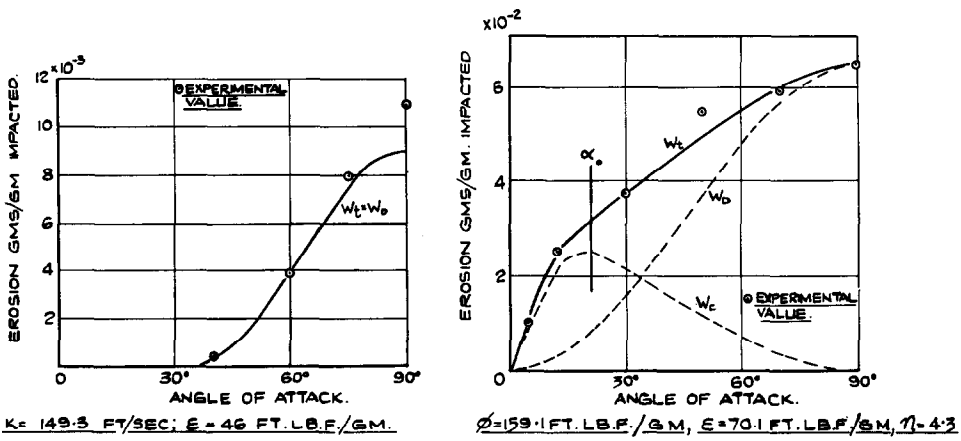


Fig. 11. Erosion *vs.* angle of attack characteristic for glass plates eroded by 475 μ diameter glass spheres at 260 ft./sec.

Fig. 12. Erosion *vs.* angle of attack characteristic for graphite plates eroded by 297 μ aluminium oxide particles at 363 ft./sec.

The characteristic for carbon plates and 297 μ angular aluminium oxide particles at 363 ft./sec is shown in Fig. 12 and the result for perspex plates and 210 μ aluminium oxide particles at 420 ft./sec is given in Fig. 13. For these cases there was again no deposition.

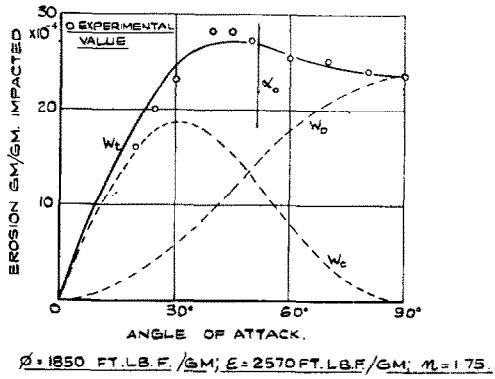


Fig. 13. Erosion *vs.* angle of attack characteristic for perspex eroded by 210 μ aluminium oxide particles at 420 ft./sec.

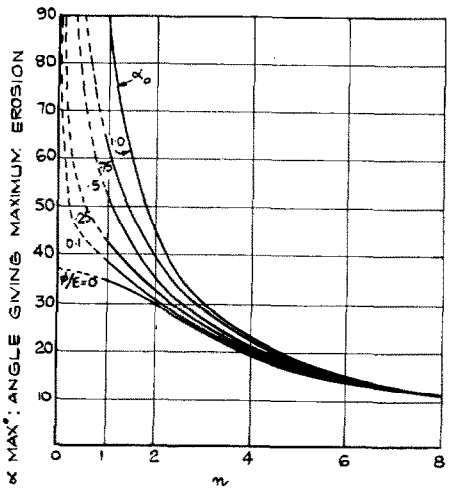


Fig. 14. Variation of α_{max} with ratio ϕ/ε and n.

EROSION RELATIONSHIPS

The experimental results confirm that two forms of erosion mechanism may act simultaneously in a given specimen under attack. For glass, which is an example of a hard brittle material, the predominant mechanism is a cracking of the surface with subsequent intersection of cracks and material removal. This appears to be associated with force components normal to the surface. For aluminium, an example of a ductile material, as well as deformation wear (work hardening due to repeated deformation then cracking of the surface layers) which accounts for erosion at normal attack, the cutting wear described by FINNIE is also in evidence. This cutting wear mechanism is associated with forces parallel to the surface under attack. For simplicity, wear due to force components normal to the surface is referred to as deformation wear for all classes of material.

The experimental results and the analyses of FINNIE and BITTER indicate that the following factors should be accounted for in any relationships for erosion damage.

(a) The normal component of kinetic energy of the impacted particles is absorbed in the specimen surface and accounts for deformation wear.

(b) For certain hard materials, subjected principally to deformation wear, there is a limiting component of velocity normal to the surface below which no erosion takes place. This limiting value is dependent on particle shape.

(c) The kinetic energy component parallel to the surface is associated with cutting wear.

(d) For cutting wear and large angles of attack the particles come to rest in the surface and the total parallel component of kinetic energy contributes to cutting wear. For small angles of attack, however, the particles may sweep into the surface and finally leave again with a residual amount of parallel kinetic energy.

If one assumes that for cutting wear ϕ units of kinetic energy must be absorbed

by the surface to release one unit mass of eroded material and that the corresponding parameter for deformation war is ε, then the above factors immediately lead to the following relationships.

$$W = \underset{(A)}{\frac{\frac{1}{2}M(V^2\cos^2\alpha - v_p^2)}{\phi}} + \underset{(B)}{\frac{\frac{1}{2}M(V\sin\alpha - K)^2}{\varepsilon}} \qquad \alpha < \alpha_0 \tag{1}$$

and

$$W = \underset{(C)}{\frac{\frac{1}{2}MV^2\cos^2\alpha}{\phi}} + \underset{(B)}{\frac{\frac{1}{2}M(V\sin\alpha - K)^2}{\varepsilon}} \qquad \alpha > \alpha_0 \tag{2}$$

where W is the erosion produced by M pounds of particles at angle of attack α and particle velocity V. K is the velocity component normal to the surface below which no erosion takes place in certain hard materials and v_p is the residual parallel component of particle velocity at small angles of attack. Part B accounts for deformation wear and parts A and C account for cutting wear at small angles of attack and large angles of attack respectively. α_0 is the angle of attack at which v_p is zero so that at this angle eqns. (1) and (2) predict the same erosion.

Now v_p will increase with decreasing angle of attack and will also depend on the initial velocity component parallel to the surface.

$$\text{Let } \gamma = \left(\frac{v_p}{V\cos\alpha}\right)^2$$

Then at $\alpha = 0$, $\gamma = 1$, and at $\alpha = \alpha_0$, $\gamma = 0$

Tests on ductile materials with constant particle velocity show that, as the angle of attack is increased from zero, the erosion initially increases at a rapid rate but at larger values of angle the rate decreases. A function of α which gives a variation similar to this is

$\gamma = 1 - \sin(n\alpha)$, where n is a constant.

Substituting this expression in the above erosion relationships gives

$$W = \underset{(A)}{\frac{\frac{1}{2}MV^2\cos^2\alpha\sin n\alpha}{\phi}} + \underset{(B)}{\frac{\frac{1}{2}M(V\sin\alpha - K)^2}{\varepsilon}} \qquad \alpha < \alpha_0 \tag{3}$$

and

$$W = \underset{(C)}{\frac{\frac{1}{2}MV^2\cos^2\alpha}{\phi}} + \underset{(B)}{\frac{\frac{1}{2}M(V\sin\alpha - K)^2}{\varepsilon}} \qquad \alpha > \alpha_0 \tag{4}$$

and when $\alpha = \alpha_0$, $\sin n\alpha_0 = 1$ *i.e.* $\alpha_0 = \dfrac{\pi}{2n}$ (5)

These equations are perfectly general and the erosion–angle of attack characteristics predicted by them for a particular material depends on the relative magnitudes of the cutting and deformation wear constants ϕ and ε. For instance, cutting wear predominates when the equations are used to describe erosion of aluminium. It is

often the case that K, appearing in term B, can be neglected as it is usually small relative to the particle velocity. For very brittle materials like glass, which cannot suffer plastic deformation, the cutting terms A and C do not apply.

Shape of the erosion–angle of attack characteristic

In the following analysis the effect of K is neglected.

For the range of angles $\alpha > \alpha_0$ differentiating eqn. (3) gives

$$\frac{dW}{d\alpha} = MV^2 \cos\alpha \sin\alpha \left(\frac{1}{\varepsilon} - \frac{1}{\phi}\right) \tag{6}$$

Three categories of $\frac{dW}{d\alpha}$ for $\alpha > \alpha_0$ are

(i) $\frac{dW}{d\alpha} = 0$; then $\varepsilon = \phi$ and $W = \frac{1}{2}MV^2/\phi$ *i.e.*, the erosion is independent of angle of attack if $\phi/\varepsilon = 1$.

(ii) $\frac{dW}{d\alpha} > 1$; then $\phi > \varepsilon$ *i.e.* the erosion increases with increasing angle of attack if $\phi/\varepsilon > 1$.

(iii) $\frac{dW}{d\alpha} < 1$; then $\phi < \varepsilon$ *i.e.* the erosion decreases with increasing angle of attack if $\phi/\varepsilon < 1$.

For the range of angles $\alpha < \alpha_0$ from eqn. (4) $dW/d\alpha = 0$ gives

$$\frac{\phi}{\varepsilon} = \sin n\alpha_{max} - \frac{n}{2} \frac{\cos n\alpha_{max}}{\tan \alpha_{max}} \tag{7}$$

where α_{max} is the angle where $dW/d\alpha = 0$ occurs.

If $\alpha_{max} = \alpha_0$ then, from eqns. (5) and (7), $\phi/\varepsilon = 1$.

If $\alpha_{max} < \alpha_0$ then eqn. (7) gives $\phi/\varepsilon < 1$.

If $\alpha_{max} > \alpha_0$ then eqn. (7) is not satisfied, no maximum occurs and $\phi/\varepsilon > 1$.

The relationship between ϕ/ε, α_{max} and n is shown in Fig. 14. For the curve of $\phi/\varepsilon = 0$, ϕ cannot be zero since this would mean infinite erosion and therefore ε is infinity implying that the deformation wear is zero. Thus for $\phi/\varepsilon = 0$ only cutting wear takes place and a typical example of a plot of eqns. (3) and (4) under these circumstances is shown in Fig. 15(a). For $\phi/\varepsilon < 1$ typical plots are shown in Fig. 15(b) and for $\phi/\varepsilon = 1$ examples are given in Fig. 15(c).

For $\phi/\varepsilon > 1$ there is no turning point and typical plots are shown in Fig. 15(d). For this type of erosion it is useful when analysing experimental data to find how ϕ/ε and n affect the angle of attack at which the erosion is half that experienced at $\alpha = 90°$. If this angle is denoted by $\alpha_{\frac{1}{2}}$ then $\alpha_{\frac{1}{2}}$ may either be greater or less than α_0.

Thus for $\alpha_{\frac{1}{2}} > \alpha_0$ eqn. (4) gives

$$\frac{\phi}{\varepsilon} = \frac{\cos^2 \alpha_{\frac{1}{2}}}{\frac{1}{2} - \sin^2 \alpha_{\frac{1}{2}}} \tag{8}$$

and for $\alpha_{\frac{1}{2}} < \alpha_0$

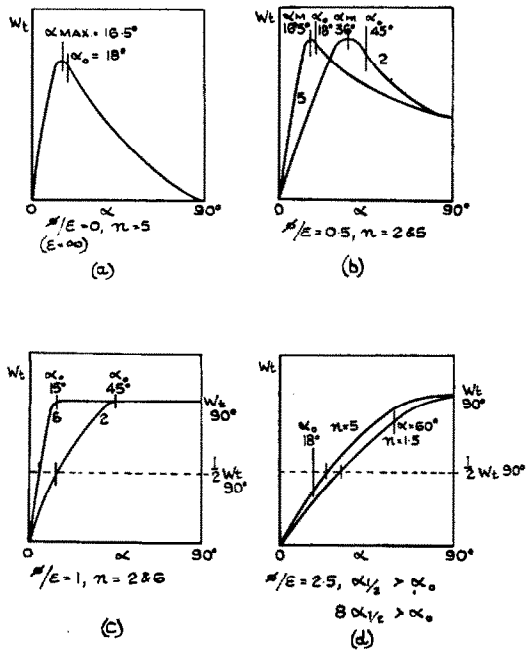


Fig. 15. Typical erosion–angle of attack characteristics.

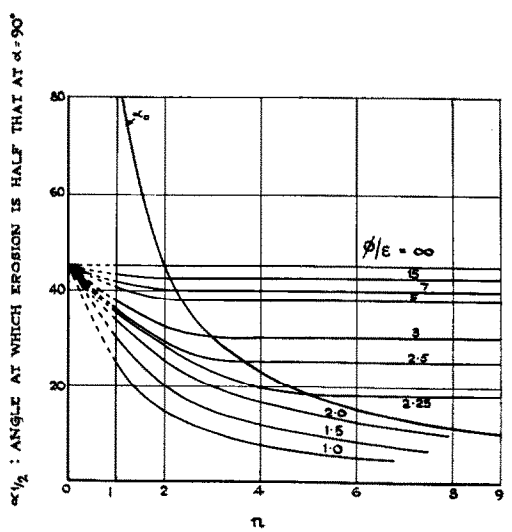


Fig. 16. Variation of $\alpha_{\frac{1}{2}}$ with ratio ϕ/ε and n.

$$\frac{\phi}{\varepsilon} = \frac{\cos^2 \alpha_{\frac{1}{2}} \sin n\alpha_{\frac{1}{2}}}{\frac{1}{2} - \sin^2 \alpha_{\frac{1}{2}}} \tag{9}$$

Equation (8) gives a direct relationship between ϕ/ε and $\alpha_{\frac{1}{2}}$ whereas in eqn. (9) the parameters are dependent on n. For convenience, the two relationships are plotted together in Fig. 16.

CORRELATION OF THE EROSION RELATIONSHIPS AND EXPERIMENTAL RESULTS

I General procedure:

From experimental results and Figs. 14 and 16, the parameters ϕ, ε and n can be obtained with the minimum of calculation. The following procedure can be adopted for any specimen material.

(i) If the actual erosion–angle of attack characteristic exhibits a maximum then ϕ/ε is less than unity.

(ii) Using the erosion value obtained at $\alpha = 90°$, ε can be obtained from $W_{90°} = \frac{1}{2} MV^2/\varepsilon$.

(iii) Using ε, the contribution that deformation wear makes to the total wear at all angles can be obtained from eqns. (3) and (4), part B.

(iv) The cutting wear can be obtained at all angles by subtracting the deformation wear from the experimental values and the angle where cutting wear is a maximum can be obtained.

(v) Using the angle for maximum cutting wear (obtained in (iv)) and the $\phi/\varepsilon = 0$ curve in Fig. 14, the values of n and α_0 can be obtained.

(vi) From Fig. 14, using the value n and the angle giving maximum erosion, the value of ϕ/ε and hence of ϕ can be obtained.

(vii) If the actual erosion–angle of attack characteristic does not exhibit a turning value then ϕ/ε is greater than unity. The procedure here is to follow (ii), (iii), (iv) and (v) above and to determine the angle giving half the erosion experienced at 90° ($\alpha_{\frac{1}{2}}$) from the actual characteristic. Thus knowing n and $\alpha_{\frac{1}{2}}$, Fig. 16 may be used to obtain ϕ/ε and ϕ. Figure 16, incidentally, also allows one to state whether $\alpha_{\frac{1}{2}}$ is greater or less than α_0.

For cases where the maximum cannot be obtained accurately it is better to find the maximum possible α_0 (greatest α_0 for the smallest n, *i.e.* when $\phi/\varepsilon = 0$) and to use a value greater than this to obtain ϕ and hence ϕ/ε. This procedure is the best one to use for ductile materials having a relatively large value of n.

II Correlation with the actual experimental results

The erosion of ductile materials

Deposition effects. In eqns. (3) and (4) no account is taken of the initial weight increase experienced by ductile materials under certain conditions. From the experimental work on aluminium specimens the following observations may be made about deposition.

(i) Effect of particle velocity. Figure 2 shows that as the velocity is increased the mass of particles required to cause a loss of weight decreases.

(ii) Effect of angle of attack. Figure 3 indicates that as the angle of attack decreases the mass of particles required to cause weight loss decreases.

(iii) No deposition is obtained with spherical particles.

(iv) Once a weight loss is detected the erosion–mass impacted relationship is linear at a given velocity and angle of attack.

(v) To allow for deposition when using eqns. (3) and (4) the slope of the erosion–mass impacted relationships for various angles of attack at a given particle velocity must be determined. These figures allow the erosion–angle of attack characteristic to be determined where the erosion is expressed as a weight loss per unit mass impacted.

Values of the parameters ϕ, ε and n. In Figs. 5, 6 and 7 the erosion–angle of attack characteristics for aluminium plates and angular particles were obtained experimentally as outlined above and were analysed to determine ϕ, ε and n using the process of Section I. The appropriate curves for cutting wear and deformation wear are shown in the figures. For results for the erosion parameters ϕ, ε and n see Table I.

TABLE I

210 μ ANGULAR ALUMINIUM OXIDE PARTICLES AND ALUMINIUM PLATES

Particle velocity (ft./sec)	ϕ *(ft.lb./g)*	ε *(ft.lb./g)*	ϕ/ε	n
354	6830	15100	0.453	5.0
424	3571	8375	0.428	5.0
494	2800	7130	0.393	4.0

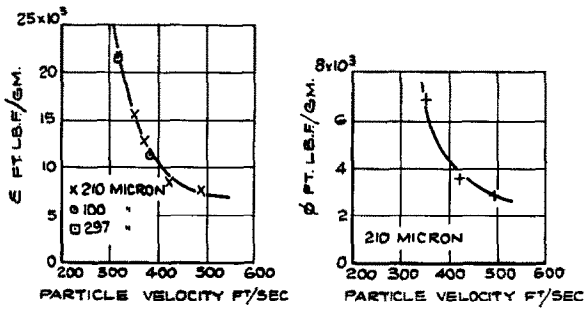


Fig. 17. Variations of cutting and deformation wear parameters with particle velocity. Aluminium plates eroded by aluminium oxide particles.

The variation of ϕ and ε with velocity for these 210 μ angular particles is shown in Fig. 17. Included in the curve for ε are values for the parameter for angular 100 and 297 aluminium oxide particles and aluminium plates obtained from tests at 90° angle of attack. These values were taken from previous work. These results indicate that ϕ and ε are velocity dependent which may be an effect of the rate of straining of the material on impact. It will be noted that there is little variation in the ratio ϕ/ε. The fact that the values of ε lie on the same curve for the three particle sizes examined suggests that particle size in itself is not a major factor affecting erosion although it is known that particle size is a major factor in the velocity attained by the particle in a given gas stream.

The much smaller erosion of aluminium plates obtained when spherical particles are used (Fig. 8) is reflected in the large values for ϕ and ε. The ratio ϕ/ε is 0.188 and, since this is considerably less than that obtained with angular particles, it would appear that the ratio ϕ/ε is dependent on particle geometry.

Materials, which have been examined and which give erosion characteristics similar to aluminium, are copper, brass and mild steel.

The erosion of very brittle materials

In a hard brittle material such as glass no deposition effects were exhibited with either spherical or angular particles. Cutting wear was negligible and with angular particles (Figs. 9 and 10) erosion was obtained at small angles of attack implying that

K in eqns. (3) and (4) was also negligible. ε was 113.4 ft.lb./g under these circumstances.

When spherical particles were used on glass plates (Fig. 11) no erosion was detected at angles less than approximately 40° and hence under these circumstances account must be taken of K. For this test K was 149.3 ft./sec and ε was 46 ft.lb./g, compared with $\varepsilon = 113.4$ for angular particles, showing that a spherical particle has a larger influence on deformation erosion for glass than an angular particle.

Other materials in this category, which have been examined and which have similar characteristics, are hard steel and various refractory materials.

The erosion of materials which are neither typically brittle nor typically ductile

From the tests on graphite (Fig. 12) and perspex (Fig. 13) it appeared that they were materials of this type, where neither cutting wear nor deformation wear predominates. For these materials no deposition effects were observed and erosion was detected at small angles of attack. Cast iron was found to have similar characteristics.

CONCLUSIONS

Particle erosion is a complex phenomenon and it is believed that the simple procedures outlined here will provide a sound basis on which other workers in this field can correlate their experimental results. There is a need for further work particularly to find how the parameters ϕ and ε vary with particle velocity. A knowledge of how the ratio ϕ/ε varies with particle velocity for different types of particle and specimen material would prove especially useful. When erosion tests are made in the laboratory, in an attempt to simulate the conditions pertaining to actual problems in which the particles are accelerated in a gas or vapour medium, great care must be taken to ensure that the particle velocity in the laboratory rig is of the same order as that in the actual case. In this respect it has been found that for an accelerating gas–particle mixture particle size, shape and density, initial particle velocity, the mass of particles per unit mass of gas, and the length of the containing duct are all important variables affecting the final particle velocity.

ACKNOWLEDGEMENTS

This work was performed at the University of Strathclyde under the general direction of Prof. A. S. T. THOMSON of the Mechanical Engineering Department. The work was initiated by the Ministry of Aviation and Imperial Metal Industries (Kynoch) Ltd., Summerfield Research Station, England.

REFERENCES

1 S. J. ROSENBERG, Resistance of steels to abrasion by sand, *Trans. Am. Soc. Steel Treating, 18* (1930) 1093.
2 A. E. SCHUH AND E. W. KERN, Measurement of abrasion resistance, *Ind. Eng. Chem. Anal. Edition, 3* (1931) 72.
3 M. A. FISHER AND E. F. DAVIES, Studies of fly-ash erosion, *Trans. A.S.M.E., 71* (1949) 481.
4 K. WELLINGER, Sandstrahlverschleiss an Metallen, *Z. Metallk., 40* (1949) 361.
5 R. L. STOKER, Erosion due to dust particles in a gas stream, *Ind. Eng. Chem., 41* (1949) 1196.
6 W. A. STAUFFER, Wear of metals by sand erosion, *Metal Prog., 69*(1) (1956) 102.
7 I. FINNIE, The mechanism of erosion of ductile metals, *Proc. 3rd. U. S. Natl. Congress Appl. Mech., 1958*, Pergamon Press, London, 1958, p. 527.
8 I. FINNIE, An experimental study of erosion, *Proc. Soc. Exptl. Stress Anal., 17* (2) (1960) 65.
9 I. FINNIE, Erosion of surfaces by solid particles, *Wear, 3* (1960) 87.
10 J. G. A. BITTER, A study of erosion phenomena, Parts 1 and 2, *Wear, 6* (1963) 5, 169.